

Mojtaba Alehosseini

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M.Sc. Computer Science (Artificial Intelligence), University of Genoa · graduating February 2027

PROFILE

Master's student in Artificial Intelligence at the University of Genoa, graduating in February 2027. Four years at a carpet and yarn manufacturer in Iran, first as systems administrator, then as the analyst who built the Power BI reporting its sales, finance and supply-chain managers work from. Co-author of a 2026 review in Advanced Healthcare Materials; co-wrote its machine-learning methodology section. Programs in Python, SQL, Java and C++, trains models in PyTorch and scikit-learn, and has run physical infrastructure: a 100-GPU fleet kept online around the clock for 28 months. Seeking a data scientist, machine learning engineer or analytics engineer role.

SKILLS

Programming: Python (pandas, NumPy, scikit-learn, PyTorch), SQL (PostgreSQL, SQLite), Java, C and C++, Bash, MATLAB, SWI-Prolog

Machine learning: classification and regression, time-series forecasting (LightGBM, SARIMA), recommender systems, rolling-origin backtesting and model evaluation, transfer learning for image classification, LLM applications (retrieval-augmented generation with ChromaDB, LoRA fine-tuning)

Systems: FastAPI, Flask, Docker, ONNX Runtime and INT8 quantisation, MLflow, DVC, GitHub Actions, Prometheus and Grafana, Linux; parallel computing with OpenMP, MPI and CUDA

Symbolic and multi-agent AI: JADE, FIPA protocols, SWI-Prolog rule bases and DCG grammars, OWL ontologies, Rasa

Business intelligence: Power BI, Power Query, Excel, KPI design, SQL data models

EXPERIENCE

Setare Talae Co. (Patris Carpet), Delijan, Iran 2021 – 2025

Carpet and yarn manufacturer. Two roles, on site and then remote.

Business Technology Analyst (part-time; remote from 2024) 2023 – 2025

- Built the Power BI reporting for sales, finance and supply chain on SQL and Power Query data models; routine reporting time fell by about 25%
- Replaced the manual spreadsheet forecast for sales and supply-chain demand with a SQL and Power Query model
- Turned managers' questions into agreed KPIs, documented each report and trained non-technical staff to use it

IT Systems Administrator 2021 – 2022

- Kept the company's servers and office network running and supported staff on site

Riggosaurus, Iran 2020 – 2025

Self-founded GPU compute operation for Ethereum mining. Co-founder and infrastructure operator.

- Built and ran 20+ open-air rigs with 100+ NVIDIA and AMD GPUs across 11 models, kept online around the clock for 28 months with remote monitoring and alerting
- Tuned each GPU model for performance per watt (clocks, voltage, memory timings) and modelled electricity, hardware and pool costs to decide what to run and when
- Sourced, tested and benchmarked second-hand GPUs; diagnosed and repaired rigs in-house

Pinfopedia, Iran 2018 – 2020

Co-founded start-up: a Persian-language product encyclopedia. Co-founder and software developer.

- Built the platform: relational database schema, Flask back end and REST API, Celery and RabbitMQ background jobs; QR-code labels linked physical products to their pages

EDUCATION

M.Sc. Computer Science (Artificial Intelligence) 2024 – Feb 2027 (expected)

University of Genoa (DIBRIS), Italy

Course projects: High Performance Computing (parallel GEMM in OpenMP, MPI and CUDA); Symbolic and Distributed AI with Natural Language Processing (Port Command Genova); Data Visualization

Top 2% of nearly 500,000 candidates in the national entrance exam (2017); 3rd in the Computer Science department (2021)

PUBLICATION

Kamaei M., Mohammadi H., Golafshan S., **Alehosseini M.**, Mohammadzadeh M., Golshirazi A., Golafshan N., Eufrásio-da-Silva T., Dolatshahi-Pirouz A., Orive G. (2026). *Intelligent Orthopedics: Machine Learning in Diagnosis of Bone Disease, Implants, and Bone Health Monitoring*. Advanced Healthcare Materials, e71447. doi.org/10.1002/adhm.71447

Peer-reviewed review article, open access. Co-wrote Section 3, Machine Learning in Orthopedics: problem definition and learning approaches, data identification and preprocessing, model selection and training, evaluation and interpretation.

PROJECTS

Port Command Genova · course project, Symbolic and Distributed AI + NLP, 2026 · GitHub: Port-Command

A single-player port-management game commanded in plain English. Ten JADE agents (harbour master, tugs, terminals, customs, weather, assistant) plus vessel agents negotiate with FIPA messages and Contract-Net auctions; every move is checked against a SWI-Prolog rule base of about 30 rules; a Prolog DCG parses negotiation sentences, with a Rasa fallback and an OWL ontology for the vocabulary; a local LLM (Phi-4-mini) explains each decision and is checked so it cannot add facts. Java 21, SWI-Prolog, Python.

Parallel dense matrix multiplication (OpenMP, MPI, CUDA) · course project, High Performance Computing, 2026 · GitHub: hpc-gemm-openmp-mpi-cuda

Double-precision GEMM as sequential (three loop orders, cache blocking), OpenMP, MPI and CUDA kernels, benchmarked on an i9-12900K, a Tesla T4 and a Tesla P100: MPI 6.47x over the best sequential kernel, CUDA on the P100 27.4x. Showed that the T4 loses to the CPU because Turing runs FP64 at 1:32, not because of the kernel. 33-page report; every figure traceable to a committed benchmark file. C, CUDA.

NL-to-SQL GenBI · personal project, 2026 · GitHub: nl-to-sql-genbi

Natural-language questions turned into read-only SQL against a live database: schema-aware prompting with retrieval over ChromaDB, read-only guardrails, automatic charts; Streamlit UI, pluggable LLMs (Ollama, Gemini, Hugging Face); 60 automated tests. Python.

Demand forecasting · personal project, 2026 · GitHub: demand-forecasting

Seasonal-naive and SARIMA baselines against LightGBM with leakage-safe features and rolling-origin backtesting; WAPE 17.0% against 43.8% for the seasonal-naive baseline on a 90-day holdout. Python.

ONNX inference service · personal project, 2026 · GitHub: onnx-inference-service

FastAPI service for a PyTorch model exported to ONNX with INT8 quantisation: 2.6x faster than PyTorch, 1.71x smaller, accuracy change of -0.0005; 24 tests; Docker. Python.

Retail recommender · personal project, 2026 · GitHub: retail-recommender

Implicit-feedback recommender (popularity, item-item collaborative filtering, matrix factorisation) on 479K purchase records from UCI Online Retail II; the next purchase lands in the top 10 for about 21% of 5.5K customers, 4x the popularity baseline. Python.

TEACHING AND SERVICE

- Teaching assistant, Yazd University: Advanced Programming (2020), Introduction to Programming (2019)
- Secretary, Computer Science Scientific Association, Yazd University, 2018 – 2020: ran programming contests, hackathons and workshops; association ranked 2nd in the university
- Persian translation of the SimPy simulation library manual (GitHub)

LANGUAGES AND WORK AUTHORISATION

Persian (native) · English (C1; TOEFL iBT 99, 2022) · Italian (A2, in progress)

Italian student residence permit valid to May 2027. Part-time work permitted now; after graduation, convertible to a work permit outside the Decreto Flussi quota (art. 22 c. 11-bis, D.Lgs. 286/1998).

I authorise the processing of my personal data in accordance with D.Lgs. 196/2003 and EU Regulation 2016/679 (GDPR).